27356

Training Institute For Chemical Industries

Polash, Narsingdi, Bangladesh

(Central Training Institute of BCIC)

Certificate

This is to certify that

Aszad Hossain Akib

has successfully completed the training course on

Quality Control and

Unit Operation in Food Industry

of 04 (Four) weeks duration conducted by this Institute

from 09-11-2024 to 05-12-2024

He/She obtained grade A *in the final evaluation.*

Date: 05 December, 2024

Executive Director

Gradation Criteria: A^+ = 80% & above A = 75 ~ <80%, A^- = 70 ~ <75%, B^+ = 65 ~ <70%, B = 60 ~ <65%, B^- = 55 ~ <60% , C^+ = 50 ~ <55%, C = 45 ~ <50%, D = 40 ~ <45%, F = Below 40%.

(Course outline overleaf)

Course Outline

Analytical Chemistry & Environmental Science (76 Hours)

Quality control (08 hours): Introduction to Quality Control & Quality Assurance; Method validation in chemical analysis; Productivity Concept in Industry; Power point presentation.

Industrial application of classical analysis (02 hours): Viscosity measurement of oil/ transparent liquid.

Analysis of industrial water (08 hours): Nanotechnology in water treatment; Preparation & standardization of solutions for titrimetric analysis; Determination of Alkalinity & Acidity in potable water; Determination of mineral content- Ca, Mg in potable water

Quality testing methods in food industries (24 hours): Estimation of Antioxidant: Ascorbic acid and pectin content in fruits; Estimation of fat in milk/ice cream/powder milk; Quantitative measurement of preservatives (E-201) in Food; Adulteration test in common food items; Estimation of Polyphenol/ carotene in fruit and vegetables; Determination of Acid number, Saponification & Peroxide value; Estimation of caffeine in tea/ coffee/ cola drink; Determination of total sugar in beverage using polarimeter; Rancidity and other parameter test of edible oil; Estimation of Alyl isothiocyanate in Mastered oil.

Spectrometric methods of analysis (16 hours): Principles and Instrumentation of Spectrophotometer; Determination of Na & K in green coconut by Flame Photometer; Identification & Quantification of analyte by UV/Vis Spectrophotometer; Estimation of vitamin-A in edible oil; Qualitative chemical analysis of trace & toxic elements by ICP-OES; Quantitative chemical analysis of trace & toxic elements by ICP-OES; Micronutrient analysis by EDXRF; Identification of characterization of graphene oxide by FTIR.

Chromatographic methods of analysis (08 hours): Chromatographic Methods of Analysis; Operation and application of Gas Chromatograph (GC); Operation and application of GC-MS; Estimation of caffeine in cola drinks by HPLC.

Pollution control in food industries (10 hours): Wastewater treatment system; Measurement of pollution parameters: DO, BOD & COD in wastewater; Operation of environmental pollution instrument.

Operation & Process Technology (18 Hours)

Unit operations in the process industry: Process symbol and process diagram; Practice on process symbol and process diagram; Operation of water treatment plant (Ion exchange); Pump: Operational behavior and characteristics; Control of process parameters using SCADA & PID; Boiler & Heat Exchanger; Solid Handling & Absorption; Evaporation & Crystallization.

Instrumentation & Control Engineering (04 Hours)

Advanced process control (basic thought): Process control devices and control loop; Process automation through PCL.

Mechanical Engineering (04 Hours)

Mechanical equipment in industry: Operation & Demonstration of refrigeration system; Demonstration of pipe fittings and valves.

Industrial Safety & Health (06 Hours)

Safety measures in the process industry: Introduction to Industrial Safety; Fire hazard & Fire protection arrangement; Explosion hazard & Personal protective equipment.

Others (04 Hours)

National Integrity Strategy (NIS); Accounts and Finance in Industrial Sector.